

Optimizing Heavy Metal Detection with Deep Neural Networks and Gaussian Mixture Clustering in Voltammetric Data

Alya Kamilah
Department of Informatics
Institut Teknologi Sepuluh Nopember
Surabaya 60111, Indonesia
6025241066@student.its.ac.id

Abdullah Faqih Septiyanto
Department of Informatics
Institut Teknologi Sepuluh Nopember
Surabaya, Indonesia
7025221022@student.its.ac.id

Fadlilatul Taufany
Department of Chemical Engineering
Institut Teknologi Sepuluh Nopember
Surabaya, Indonesia
f_taufany@chem-eng.its.ac.id

Riyanarto Sarno
Department of Informatics
Institut Teknologi Sepuluh Nopember
Surabaya 60111, Indonesia
riyanarto@its.ac.id

Rizqy Ahsana Putri
Department of Informatics
Institut Teknologi Sepuluh Nopember
Surabaya 60111, Indonesia
6025231069@student.its.ac.id

Dwi Sunaryono
Department of Informatics
Institut Teknologi Sepuluh Nopember
Surabaya 60111, Indonesia
dwi@its.ac.id

Kelly Rossa Sungkono
Department of Informatics
Institut Teknologi Sepuluh Nopember
Surabaya 60111, Indonesia
kelly@its.ac.id

Taufiq Choirul Amri
Department of Informatics
Institut Teknologi Sepuluh Nopember
Surabaya, Indonesia
7025222001@student.its.ac.id

Shoffi Izza Sabilla
Department of Medical Technology
Institut Teknologi Sepuluh Nopember
Surabaya, Indonesia
shoffi@its.ac.id

Abstract— The accurate and efficient detection of heavy metal is of paramount importance, given their deleterious effects on human health and the environment. While traditional laboratory methods are highly accurate, they are not without challenges, including high cost, non-portability, and time-consuming procedures. This study introduces a novel framework that integrates deep learning and clustering techniques to enhance the detection and classification of heavy metal from voltammetric data. The data were collected using cyclic voltammetry (CV) methods with a glassy carbon electrode (GCE) and subsequently analyzed using five classifiers. The following classifiers were employed: Support Vector Machine (SVM), Random Forest, K-Nearest Neighbors (KNN), XGBoost, and Deep Neural Networks (DNN). A dataset comprising 750 samples, categorized into three groups (cadmium, non-heavy metal, and lead), was subjected to advanced preprocessing techniques, including data normalization and dimensionality reduction through principal component analysis (PCA). The DNN model demonstrated superior accuracy, achieving 99.87%. This highlights its capacity to process noisy and complex data more effectively than traditional methods. Additionally, Gaussian Mixture Models (GMM) were employed to cluster and identify anomalies within the data. Despite its high accuracy, this study discusses limitations such as computational complexity and potential challenges in highly contaminated environments. These results underscore the significance of developing portable, efficient detection systems for on-site applications, offering a notable advancement over traditional techniques.

Keywords— *Deep Learning, Heavy Metal Detection, Machine Learning, Voltammetric*

I. INTRODUCTION

Metallic elements classified as heavy metal have a high specific density relative to water, typically exceeding 5 g/cm³ (1). A number of heavy metal, including Pb, Zn, Cu, Ag, Fe, Hg, Cr, As, Cd, Ni, Pd, U, and Pt, have been identified as having the potential to cause harm and can negatively impact a number of health aspects, even at low exposure levels (2, 3, 4). The introduction of heavy metal into the human body can

result in the disruption of biological functions, leading to the development of long-term health complications. These include damage to vital organs and the nervous system. It is therefore imperative to be able to detect the presence of heavy metal in the environment, which makes this research a matter of great urgency in the fields of environmental science and public health.

The principal danger posed by heavy metal arises from their capacity to accumulate in biological tissues, thereby exerting toxic effects. Exposure to heavy metal can result from natural processes or human activities (anthropogenic sources) such as industrial processes, agriculture, pharmaceuticals, waste disposal, and mining. The increase in these activities raises the risk of heavy metal contamination, which harms the atmosphere, soil, and groundwater and can subsequently enter the human food chain. To address the widespread impact of such contamination, accurate identification of heavy metal types and concentrations in the environment is essential.

While heavy metal detection can be accomplished through laboratory-based instruments, these tools are typically non-portable, costly, and necessitate the expertise of trained operators [9]. The use of advanced laboratory methods is often limited to stationary facilities, which makes them unsuitable for field applications. It is therefore imperative to develop devices that are more affordable, efficient, user-friendly, and accurate, particularly for emergency field use in detecting heavy metal.

Voltammetry is an electrochemical technique that is commonly employed for the purpose of testing heavy metal ion concentrations. This technique relies on a three-electrode potentiostat sensor to capture current signals corresponding to the concentration of the analyte being tested. Prior research has demonstrated the effectiveness of voltammetry in identifying the type and concentration of heavy metal ions. However, despite these advances, voltammetry still faces challenges in efficient data collection and optimization for

field measurements. Consequently, the development of automated data processing methods is critical to addressing limitations in voltammetry.

Conventional methodologies for the processing of heavy metal data frequently entail manual techniques, such as square-wave voltammetry, wherein polar analysis facilitates the differentiation of ion species and concentrations through the observation of distinct redox peaks [12]. The manual processing of data necessitates the conventional comparison of the half-wave potential of the standard solution with that of the tested solution. Although this method is effective, it is time-consuming and inefficient for rapid analysis. Related research in heavy metal detection has demonstrated the efficacy of machine learning-based data processing. However, equipment constraints and labor requirements persist as obstacles to optimization and efficiency in field analysis.

The objective of this research is to develop a more efficient heavy metal detection system by employing a combination of machine learning and deep learning models. The proposed system is designed to achieve greater accuracy in the detection of heavy metal in a shorter timeframe. The machine learning models employed in this study include Support Vector Machine (SVM), Random Forest, K-Nearest Neighbors (KNN), XGBoost, and Deep Neural Network (DNN). The preliminary findings indicate that the DNN model exhibits the most optimal performance among these models in classifying heavy metal. Furthermore, the method evaluation is conducted using classification matrices, classification reports, and performance analysis through precision, recall, and F1-score metrics.

II. RESEARCH METHOD

A. Voltammetric Data

An electrochemical method known as cyclic voltammetry (CV) is often employed to examine complexation mechanisms, particularly in the field of environmental chemistry, where it is used to gain insight into the binding of heavy metal to organic ligands. In this investigation, a Glossy Carbon Electrode (GCE) was utilized as the working electrode, and a sintered Reference Electrode (RE) was used to maintain potential stability, producing voltammograms that illustrate the relationship between current and potential and provide insights into electrochemical reactions within the sample.

This study aims to develop an efficient system for detecting heavy metal using a combination of machine learning and deep learning models. The proposed system integrates Support Vector Machine (SVM), Random Forest, K-Nearest Neighbors (KNN), XGBoost, and Deep Neural Network (DNN) models. Initial tests indicate that DNN outperforms other models in classifying heavy metal accurately. Model evaluation includes classification matrices, precision, recall, and F1-scores, which support the development of a faster, more accurate field detection system.

B. Random Forest

Random Forest is an ensemble learning model composed of multiple independent decision trees. A tree structure is automatically generated for data distribution according to the same classification [17]. It leverages the bagging technique to

enhance both stability and predictive accuracy. The primary equation of Random Forest is to combine predictions from multiple decision trees, thereby reducing overfitting while improving accuracy. This approach is particularly effective for datasets with high variability, where individual models may struggle to generalize.

The Random forest Shown in Eq (1) :

$$H(x) = \text{mode}\{h_1(x), h_2(x), \dots, h_M(x)\} \quad (1)$$

Where :

$h_1(x)$: prediction from the i -th tree

M : denotes the total number of trees.

Equation (1) aggregates predictions by combining the outputs from multiple trees, effectively reducing the probability of prediction errors. In this study, Random Forest is applied to detect anomalies in voltammetry data, capitalizing on the model's capability to manage complex and noisy data. Additionally, this model can determine the most significant features in classifying heavy metal.

C. Support Vector Machine (SVM)

Support Vector Machine (SVM) is a classification algorithm that works by finding an optimal hyperplane to separate data into two or more classes. For multi-class classification, SVM typically uses a One-vs-One or One-vs-Rest approach. The algorithm aims to maximize the margin between two classes, where a larger margin results in better generalization of the model to unseen data. SVM is particularly effective for high-dimensional datasets and in scenarios where the number of samples is smaller than the number of features due to its ability to address overfitting issues. The decision equation can be expressed as:

$$f(x) = \text{sign}(w^T x + b) \quad (2)$$

Where:

$f(x)$: decision equation

w : weight vector, determines hyperplane orientation.

b : bias, shifts the hyperplane.

The SVM shown in Eq (2) is used to separate data into different classes with maximum margin, which helps enhance the model's generalization ability. In this research on heavy metal detection, SVM is employed to distinguish classes such as Cadmium, Non-Heavy metal, and Lead based on voltammetry features. By maximizing the margin between classes, SVM can minimize classification errors.

D. K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a classification algorithm that groups data based on the distance to its nearest Neighbors. This is done on the assumption that data with similar features are likely to belong to the same class[19]. KNN operates by identifying the closest data points (neighbors) to a given sample and assigning the majority class among those neighbors as the predicted class. This algorithm is particularly effective with datasets that are relatively clean and have a moderate number of features, as a high number of features can increase computational complexity and decrease accuracy. KNN is also sensitive to the scale of the data, so

preprocessing steps like normalization or standardization can significantly improve its performance. The KNN value for a data point is defined as:

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^n (x_{ik} - x_{jk})^2} \quad (3)$$

Shown in Eq (3) is used to find the nearest Neighbors, which are then utilized to determine the class of new data. In this study, KNN can be applied to detect heavy metal based on the distance between existing voltammetry samples. The model will predict the class of a new sample based on similar samples in the training dataset.

E. XGBoost

XGBoost is a gradient boosting-based ensemble model known for its speed and performance. It constructs decision trees sequentially, with each tree correcting the errors of the previous one. This iterative process helps build increasingly complex models, enhancing prediction accuracy. Due to its high efficiency and accuracy, XGBoost is widely used in data science competitions. The XGBoost equation minimizes the loss equation through XGBoost is reliable for this research due to its ability to quickly and efficiently handle large and complex voltammetry datasets. This model can assist in detecting heavy metal with high accuracy.

F. Deep Neural Networks

A Deep Neural Network (DNN) is an artificial neural network with multiple hidden layers that enables the model to recognize complex patterns in data [18]. The DNN is used to perform classification based on the processed data, utilizing cross-validation (Stratified KFold) to ensure thorough testing of the model. Each layer of neurons in the DNN processes the data using the equation:

$$a^{(l)} = \sigma(w^{(l)} \cdot a^{(l-1)} + b^{(l)}) \quad (4)$$

where:

- $a^{(l)}$: activation in the l -th layer
- $w^{(l)}$: weight in the l -th layer
- $b^{(l)}$: bias in the l -th layer
- σ : activation equation

Shown in Eq (4) is used to calculate the output of each neuron, which is ultimately applied to classify the data. In this study, the DNN is utilized for the final classification of voltammetry data. By employing cross-validation, the model is tested to ensure prediction accuracy in detecting types of heavy metal. This model is designed to learn complex patterns within voltammetry data, enabling high-accuracy detection of heavy metal.

G. Dimensionality reduction and clustering

A Principal Component Analysis (PCA) was conducted to reduce the dimensionality of the dataset, thereby facilitating data visualisation and analysis. Two principal components were retained in order to maintain the majority of the variance

within the data set, while avoiding a significant loss of PCA is a method of ensuring computational efficiency while maintaining the integrity of the dataset for subsequent classification tasks. The selection of two components is supported by the fact that these components are capable of highlighting the separability between classes without a significant increase in computational overhead.

Subsequently, Gaussian Mixture Models (GMM) was applied to the PCA-transformed data in order to cluster it into three groups, which represented the three heavy metal classes. The probabilistic approach employed by GMM effectively captures multivariate distributions, thereby enabling the identification of patterns and outliers within the data. The results of the clustering process provided insights regarding the separability of the aforementioned classes, thus supporting the evaluation of classification models.

III. PROPOSED METHOD

The objective of the methodology presented in this study is to distinguish between valid and invalid data. The latter is illustrated in Fig. 1.

A. Data Preprocessing

The dataset in this study includes three classes: Cadmium, Non-Heavy metal, and Lead, with 250 samples each. Relevant columns are selected to ensure consistency, and data from each class is merged into a single DataFrame. After removing entries with missing values, the data is converted to a NumPy array, standardized with StandardScaler, and normalized to prevent bias from feature scale differences.

This study assesses the classification accuracy of machine learning models DNN, SVM, XGBoost, Random Forest, and KNN in detecting heavy metal in voltammetric data across the three classes.

B. DNN-GMM Integration for Heavy Metal Detection

The present study proposes a novel combination of deep neural networks (DNN) and Gaussian mixture models (GMM) for the detection and classification of heavy metals in voltammetry data. In contrast to the majority of machine learning methods, which typically necessitate manual feature extraction, the proposed DNN in this study directly processes raw voltammetric signals to identify intricate patterns. GMM further enhances this approach by providing a probabilistic framework for clustering and outlier detection, thereby reinforcing the robustness of the system.

Fig.2 GMM performs clustering on the PCA-transformed data. GMM groups the data into three clusters, labeled 0, 1, and 2, according to the distribution characteristics of the data. The selection of two principal components by PCA offers computational efficiency, while retaining 95% of the variance, thus preparing it for further clustering. These results demonstrate that the combination of PCA and GMM effectively identifies patterns within the data, separating it into three main groups corresponding to the three classes in this study: Cadmium, Non-Heavy metal, and Lead.

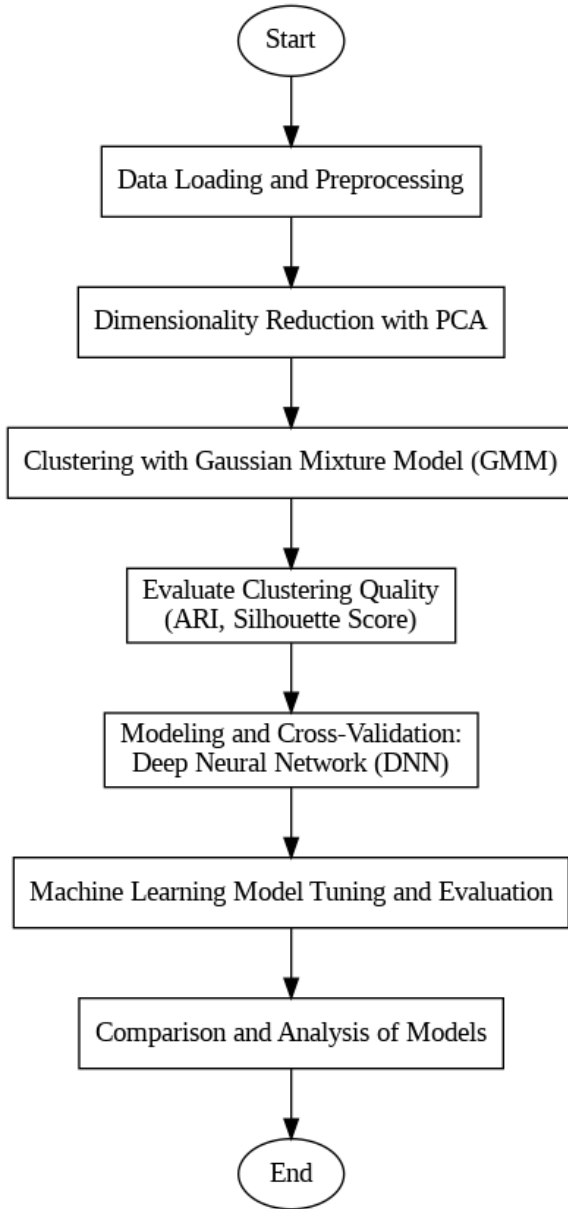


Fig. 1. Proposed method

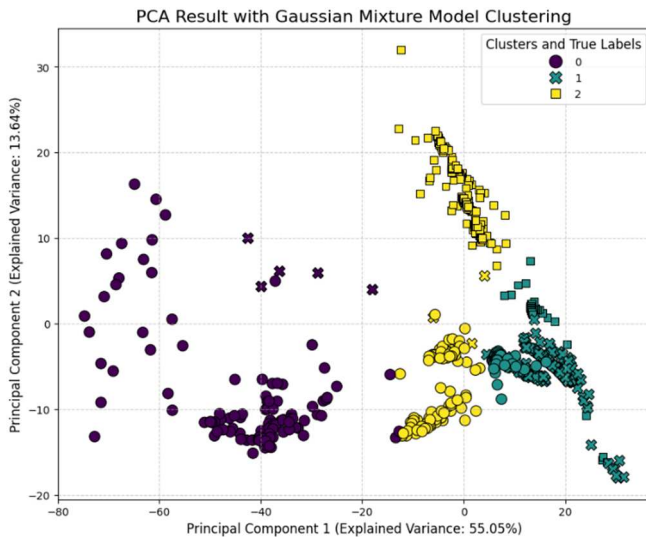


Fig. 2. PCA result with GMM

TABLE I MODEL ACCURACY

Model	Accuracy	Precision	Recall	F1-Score
Support Vector Machine (SVM)	88%	0.88	0.88	0.88
Random Forest	83%	0.86	0.83	0.83
K-Nearest Neighbors (KNN)	84%	0.87	0.84	0.84
XGBoost	88%	0.88	0.88	0.88
Deep Neural Network (DNN)	99.87%	1.00	1.00	1.00

IV. EXPERIMENTAL RESULT

A. Evaluation

The evaluation of classification results using various machine learning models shows that DNN achieved the best performance with a perfect accuracy score 99.87% across all metrics TABLE I, indicating high effectiveness in classifying heavy metal (cadmium, non-heavy metal, and lead) in voltammetric data. In contrast, Support Vector Machine (SVM), after tuning the model demonstrated a commendable degree of accuracy, reaching 88% precision, a notable enhancement from its previous accuracy of 85%. Furthermore, the distribution of precision, recall, and F1-score across all classes exhibited greater uniformity.

XGBoost also demonstrated strong performance with an accuracy of 88% after tuning, showing enhanced precision and recall in each class, particularly for the non-heavy metal class, resulting in a balanced and competitive performance. Random Forest, on the other hand, achieved an accuracy of 83% post-tuning, with a slight improvement in recall for the cadmium class, leading to better overall balance compared to its initial accuracy of 82%.

K-Nearest Neighbors (KNN) achieved an accuracy of 84% after tuning, with minor improvements in precision and recall across all classes. Meanwhile, Decision Tree increased to 83% after tuning, with enhanced precision and recall, particularly in the cadmium and lead classes. Initially, this model had an accuracy of 79%, underscoring the importance of parameter tuning in improving its performance.

The inferior performance of KNN and Random Forest relative to DNN may be attributed to their inability to effectively extract complex non-linear patterns from voltammetric data. Furthermore, although DNN offers the highest accuracy, it requires a significantly greater computational resource compared to SVM and XGBoost, which are more efficient for real-time applications. Further studies could investigate the potential of ensemble models or hybrid techniques that integrate the capabilities of DNN with those of decision tree-based methods, such as XGBoost.

The high accuracy observed in the DNN model may be attributed to overfitting on specific patterns in the training data, which could limit its robustness in datasets with a high proportion of outliers. Additionally, the model's reliance on predefined classes may limit its generalisability in environments with mixed or unknown heavy metal compositions.

The results demonstrate that DNN is the optimal choice for this categorization activity, which offers unparalleled precision. However, SVM and XGBoost also provide competitive performance with an accuracy of 88% post-tuning, making them viable alternatives when computational efficiency is a consideration.

V. CONCLUSIONS

This study proposes a novel method approach to the application of state-of-the-art machine learning methods for the accurate detection of heavy metals in voltammetry data. The approach entails the utilisation of a multitude of machine learning models, including Support Vector Machines (SVM), Random Forest, K-Nearest Neighbours (KNN), XGBoost, and Deep Neural Networks (DNN). The data processing stage comprises three main steps: data normalisation, the removal of incomplete data, and the application of Principal Component Analysis (PCA) for the reduction of dimensions. A Gaussian Mixture Model (GMM) is employed for the purpose of clustering.

The results demonstrate that the DNN model exhibited the most effective performance in classifying heavy metals, including cadmium and lead, as well as non-heavy metals, within the context of voltammetric data. Indeed, the evaluation results demonstrate that it performed its task with precision and excellence across all presented evaluation criteria. Meanwhile, SVM and XGBoost demonstrated comparable accuracy following tuning, positioning them as viable alternatives, particularly in terms of computational efficiency.

This study has presented a basic framework for the integration of machine learning and deep learning techniques in heavy metal detection, addressing some critical challenges in environmental monitoring and developing towards practical, portable solutions. Further improvements in accuracy and efficiency may be achieved by trialling other deep learning models or incorporating other approaches, such as transfer learning, in future studies. Additionally, the implementation of a portable device-based detection system may be explored to develop more practical field-ready applications.

ACKNOWLEDGMENT

This research was funded by Institut Teknologi Sepuluh Nopember (ITS) under Penelitian Kolaborasi Pusat and under the Project Scheme of the Publication Writing and Intellectual Property Rights (IPR) Incentive (Penulisan Publikasi dan Hak Kekayaan Intelektual/PPHKI) Program 2024

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